

A Prototype System for Real-Time Predictive MIDI Feedback in Improvised Performance

ABSTRACT

Improvised performance is inherently adaptive, requiring musicians to continuously respond to evolving sonic and contextual cues. Existing algorithmic composition and interactive music systems provide generative outputs, yet few explore predictive, real-time MIDI feedback that simultaneously integrates audio features and performer decisions. This paper presents a prototype system designed to capture synchronised high-resolution audio and MIDI data, analyse temporal and spectral features, and provide machine-learning-informed feedback to create real-time parameter adjustments on MIDI-controllable devices. Emphasising interactivity rather than automation, the system preserves performer agency, allowing emergent, evolving improvisational structures. We describe the system architecture, data collection and preprocessing, initial modelling approaches, evaluation metrics, and related work, demonstrating a proof-of-concept performance framework and outlining directions for future ensemble-scale and real-time applications.

1. INTRODUCTION

Improvised music performance demands continuous, context-sensitive decision-making, where performers (often subconsciously) negotiate interactions between their instrument, other musicians, and the wider acoustic environment. Traditional algorithmic composition frameworks and interactive systems [2], [6] often operate in pre-determined or symbolic domains, limiting their responsiveness to real-time performance nuances. In contrast, the system we present investigates predictive, real-time MIDI assistance that integrates audio and control data streams and provides feedback that leaves final artistic choices to the performer.

The system captures high-resolution MIDI messages, including note-on/off events, control changes, and program changes, alongside audio features such as RMS, spectral centroid, spectral flux, spectral roll-off, and frequency band energies. Predictive models analyse these streams to estimate likely next MIDI events that could take place, (for example suggesting EQ adjustments to a MIDI controlled Source Audio EQ2 pedal, or sound modulation changes on various effects devices). By maintaining a human-in-the-loop paradigm, the system supports performer agency, allowing emergent gestures to

shape real-time improvisation rather than enforcing predetermined sequences.

This paper documents the prototype architecture, experimental methodology, and early evaluation of predictive modelling strategies, establishing a proof-of-concept for ML-supported improvisation and laying the groundwork for future multi-performer ensemble scenarios.

The related code for this paper can be found at: <https://github.com/anonymised-for-review>

2. PROTOTYPE SYSTEM DESIGN

2.1. Hardware Integration

The system design integrated multiple hardware components to support real-time feedback. The performer's instrument, in this case guitar, was routed through multiple MIDI-enabled effects, including modulation, delay, loopers and EQ processors. All MIDI IN and MIDI OUT messages were then routed through a Morningstar MC8 MIDI controller and the resulting data stream was logged. Audio output was also captured through an audio interface, enabling high-resolution spectral and temporal analysis.

MIDI events, encompassing note-on/off messages, control changes, program changes, MIDI channel changes, etc. were timestamped and synchronised with extracted audio features. This dual-stream approach ensured that both the performer decisions being made and the resulting sound characteristics were made available for predictive modelling.

2.2. Data Logging and Database

For the purpose of the proof-of-concept system, data was logged into an SQL database which offered structured and query-based access for model training and evaluation.

The database schema captures:

- **Audio features:** RMS, RMS delta, spectral centroid, spectral roll-off, spectral flux, band energies (low, mid, high), and onset strength.
- **MIDI related events:** channel, note, velocity, CC number and value, program number, and event type.

The initial structured format of this system supports high-fidelity temporal analysis and enables sequential modelling for predictive tasks. However, it should be noted that a more sophisticated approach for data

collection and storage will be needed as data volume increases.

2.3 Software and Feature Extraction

The prototype system was implemented in Python, utilising standard libraries for real-time audio and message capture. Audio feature extraction employed spectral analysis techniques [9], providing descriptors of timbre, dynamics, and transient behaviors. MIDI events were captured concurrently, together creating a synchronised dataset that reflected both sound characteristics and performer parameter changes.

The core principle of the system is interactivity rather than automation. The aim is that the performer should be able to set the amount and intensity of predictive feedback from the system, ensuring that human creativity remains central while benefiting from data-informed guidance. These performer preferences can also be modelled over time using the data from multiple performances.

3. MACHINE LEARNING METHODOLOGY

3.1 Sequential Modelling

The machine learning component of the system was designed to handle time-series prediction, leveraging the temporal structure inherent in improvised performance. Models operate on sliding windows of recent audio and MIDI history to anticipate likely next events. Early experiments have focused on predicting control change adjustments for devices such as EQs, guided by the evolving spectral content of the audio signal.

To capture sequential dependencies, the prototype employed Long Short-Term Memory (LSTM) networks, a class of recurrent neural networks (RNNs) capable of learning long-range temporal relationships in sequences [4]. Windows of 500–2000 milliseconds of prior audio and MIDI states provided the contextual basis for predicting the next MIDI events, including control change number, control change value, and MIDI channel. The sequential modelling approach allowed the system to anticipate performer parameter changes and suggest contextually appropriate parameter changes in response, rather than relying solely on instantaneous feature values.

While other sequence models such as Temporal Convolutional Networks (TCNs) were considered for their computational efficiency and low-latency receptive fields, the current prototype has focused mostly on LSTMs to establish a proof-of-concept sequential prediction before exploring alternative architectures.

3.2 Feature Engineering

Predictive features integrate both audio descriptors and MIDI state information, creating a multi-modal representation of performance context. The features included:

- **Audio vectors:** RMS, RMS delta, spectral centroid, spectral flux, spectral roll-off, band energies (low, mid, high), and onset strength.
- **MIDI state:** Note-on/off events, control change values, velocities, program changes, and channel information.

The targets for prediction were the next MIDI events, specifically the upcoming channel, control change number, and control value. Audio features were normalised and categorical MIDI fields such as event type were label encoded to ensure stable model training. This feature engineering approach ensured that both short-term temporal fluctuations and the overall state of the performance are captured for predictive modelling.

3.3 Model Classes

Three model classes were explored in the prototype:

3.3.1 LSTM (Long Short-Term Memory networks)

LSTMs are ideal in that they can capture sequential dependencies in musical performance, learning temporal patterns across sliding windows of audio and MIDI history. They are particularly suited for predicting control changes and note sequences over time, though they require longer training periods and careful tuning to balance responsiveness with stability.

3.3.2 Random Forests

The choice of this model was made as it offered rapid prototyping and interpretable insights into feature importance, allowing quick evaluation of how audio and MIDI features might contribute to predictions. However, its utility is limited, as it does not inherently model sequential dependencies, relying instead on snapshot data at each timestep.

3.3.3 Gradient Boosted Trees

This model was chosen as an extension to the Random Forest model whereby the model can iteratively improve predictions by minimising error across the model ensemble. Gradient boosting provided enhanced predictive accuracy and feature insight compared to Random Forest but shared the same limitation of lacking temporal awareness.

Taking these three approaches allowed the examination of trade-offs between temporal sensitivity, predictive accuracy, and interpretability, providing a foundation for selecting models that might best support real-time music improvisation.

4. PROTOTYPE EVALUATION

Evaluation of the system focused on both technical feasibility and performer experience, reflecting the system's dual aim of algorithmic support and human-in-the-loop improvisation. Given the limited dataset

collected to date, all assessments are preliminary, intended to establish proof-of-concept rather than definitive performance metrics.

4.1 Technical Assessment

Preliminary offline evaluation provides a first indication of model behavior when predicting MIDI events from combined audio and control data. Metrics at this stage are intended to assess feasibility and plausibility and may change as the project evolves.

Key quantitative measures include:

- **Mean Squared Error (MSE):** this metric evaluates continuous parameters such as control change values, indicating how closely predicted values align with recorded performance data.
- **Classification Accuracy:** this metric assesses discrete events, including MIDI channel and control change number, representing the proportion of correctly predicted events.
- **Temporal Coherence:** this metric measures the sequential plausibility of predicted events. LSTM models capture temporal dependencies directly, while tree-based models (Random Forests and Gradient Boosted Trees) rely on snapshot-based features, limiting long-term sequential modelling.

These initial metrics have been used to provide a baseline for algorithmic performance, allowing early comparison of different model classes and helping to identify feasible configurations for real-time interaction. It is important to note that technical accuracy alone does not determine musical value; even imperfect predictions can spark new improvisational directions, reinforcing the exploratory nature of the prototype.

4.2 Performer-Centred Evaluation

Central to the system is the performer experience, emphasising interactivity and creative agency. Preliminary evaluation involved live interaction, where performers responded to system-generated suggestions in real time, with observations recorded qualitatively.

Key dimensions of this assessment will include:

- **Musicality:** Do predicted events inspire or complement improvisational choices?
- **Usefulness:** Are suggestions contextually relevant to the performer's intent?
- **Responsiveness:** Does the system react in a musically appropriate timeframe?
- **Inspiration:** Does interaction with the system encourage exploration of new sonic possibilities?

Initial observations suggest that even predictions with modest accuracy can generate meaningful musical

outcomes, demonstrating that performer creativity can leverage algorithmic guidance in unpredictable and productive ways.

Given the subjective nature of this evaluation, future work will need to incorporate structured feedback mechanisms, such as performer surveys, or real-time annotations of system interventions. These methods will allow more systematic comparison across models, sessions, and different performers, while preserving a human-centred approach.

4.3 Observations

Together, the preliminary technical and performer-centred assessments indicated that:

1. Sequential models (LSTMs) provide the most coherent temporal predictions, supporting contextually appropriate suggestions, albeit with longer training times.
2. Tree-based models (Random Forests and Gradient Boosted Trees) offer rapid prototyping and interpretability, useful for early-stage exploration, but lack inherent sequential awareness.
3. Musical utility is not fully captured by technical metrics alone; subjective performer feedback is crucial for assessing the system's creative impact.
4. The prototype successfully demonstrates the feasibility of real-time, predictive MIDI feedback, forming a foundation for further data collection, model refinement, and structured evaluation in live improvisation settings.

5. RELATED WORK

The use of interactive systems for music performance has a long history. Adaptive accompaniment systems [7] and live coding frameworks [5] primarily target symbolic music or pre-programmed behaviours. Real-time audio analysis for adaptive effects has also been explored [8]. However, these approaches rarely integrate predictive MIDI feedback in a unified framework.

Recent work in machine learning for music focuses on symbolic sequence generation, such as LSTMs for MIDI [3] or reinforcement learning for expressive performance [1]. The approach taken here extends this paradigm by integrating high-resolution audio features with MIDI histories to produce actionable, real-time suggestions, maintaining a human-in-the-loop approach that differs from fully generative systems.

6. DISCUSSION

This prototype demonstrates that integrating audio features with MIDI histories can enhance predictive modelling, improving both the quality and contextual relevance of system-generated suggestions. A central insight from this work is that human agency remains

paramount: the system is designed to provide guidance rather than control, ensuring that creative decisions remain firmly in the hands of the performer. Additionally, temporal modelling proves critical; sequence-based models such as LSTMs capture musical phrasing and sequential dependencies, resulting in predictions that are more coherent and musically plausible. The prototyping process has also shown the value of exploring multiple modelling approaches. Tree-based models offer rapid insight and interpretability, supporting early evaluation, while recurrent networks enable expressive, temporally aware predictions that align more closely with the evolving structure of improvisational practice.

The current system also has limitations. Evaluation has been confined to a single-performer context, latency remains at prototype levels, and model complexity is constrained by the available dataset. Future work will address these limitations by exploring multi-performer ensembles, implementing adaptive real-time learning, and expanding predictive targets to encompass a wider range of musical parameters and interactive possibilities.

7. CONCLUSION AND FUTURE WORK

This paper has presented a prototype system that can act as a framework for real-time, predictive MIDI feedback in improvised performance. By integrating audio analysis, high-resolution MIDI capture, and sequential machine learning models, the system illustrates how predictive suggestions can remain performer-driven, supporting emergent and evolving improvisational gestures rather than dictating outcomes. Evaluation has considered both technical performance metrics and the subjective experience of the performer, highlighting that even predictions with modest accuracy can inspire creative exploration and novel musical directions. While this work is preliminary, limited to a single performer, prototype-level latency, and constrained model complexity, it lays the groundwork for future investigations into multi-performer ensembles, adaptive real-time effects control, and more sophisticated predictive strategies. In doing so, the paper emphasises the potential of human-AI co-creativity, where algorithmic guidance complements, rather than replaces, the expressive agency of the performer.

8. REFERENCES

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